ERTH555 Problem Set 3

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Problem 1

Here you will explore why ϵ_{ij} is associated with strain and ω_{ij} is associated with rotation.

a. First write out the components of ϵ_{ij} and ω_{ij} in two dimensions (2D).

Using equation for ϵ and ω, we can write out their components as:

$$\epsilon_{11} = \frac{1}{2}(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_1}{\partial x_1}) = \frac{\partial u_1}{\partial x_1}, \tag{1}$$

$$\epsilon_{12} = \frac{1}{2}(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}), \tag{2}$$

$$\epsilon_{21} = \frac{1}{2}(\frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_2}) = \epsilon_{12}, \tag{3}$$

$$\epsilon_{22} = \frac{1}{2}(\frac{\partial u_2}{\partial x_2} + \frac{\partial u_2}{\partial x_2}) = \frac{\partial u_2}{\partial x_2} \tag{4}$$

and

$$\omega_{11} = \frac{1}{2}(\frac{\partial u_1}{\partial x_1} - \frac{\partial u_1}{\partial x_1}) = 0, \tag{5}$$

$$\omega_{12} = \frac{1}{2}(\frac{\partial u_1}{\partial x_2} - \frac{\partial u_2}{\partial x_1}), \tag{6}$$

$$\omega_{21} = \frac{1}{2}(\frac{\partial u_2}{\partial x_1} - \frac{\partial u_1}{\partial x_2}) = -\omega_{12}, \tag{7}$$

$$\epsilon_{22} = \frac{1}{2}(\frac{\partial u_2}{\partial x_2} - \frac{\partial u_2}{\partial x_2}) = 0 \tag{8}$$

where each partial derivative is evaluated at $\bar{x_0}$.

b. Assuming no rigid-body translation, i.e. $u_i(\bar{x_0}) = 0$, use equation (1) to write out the two components of displacement in terms of $\epsilon_{ij}$and ω_{ij}.

Equation (1) states:

$$u_i(\bar{x}) = u_i(\bar{x_0}) + \epsilon_{ij}dx_j + \omega_{ij}dx_j \tag{9}$$

Expanding the sums for the 1st component of u gives:

$$\begin{aligned} u_1(\bar{x}) &= u_1(\bar{x_0}) + \epsilon_{1j}dx_j + \omega_{1j}dx_j && (10)\\ &= 0 + \epsilon_{11}dx_1 + \epsilon_{12}dx_2 + \omega_{11}dx_1 + \omega_{12}dx_2 && (11)\\ &= \epsilon_{11}dx_1 + \epsilon_{12}dx_2 + \omega_{11}dx_1 + \omega_{12}dx_2. && (12) \end{aligned}$$

Doing the same for the 2nd component yields:

$$\begin{aligned} u_2(\bar{x}) &= u_2(\bar{x_0}) + \epsilon_{2j}dx_j + \omega_{2j}dx_j && (13)\\ &= 0 + \epsilon_{21}dx_1 + \epsilon_{22}dx_2 + \omega_{21}dx_1 + \omega_{22}dx_2 && (14)\\ &= \epsilon_{21}dx_1 + \epsilon_{22}dx_2 + \omega_{21}dx_1 + \omega_{22}dx_2. && (15) \end{aligned}$$

c. Substitute your results from part (a) into part (b) so that you write the two components of displacement in terms of the displacement gradients.

Making the substitution for the ϵ_{ij} and ω_{ij} terms gives the following expression for the 1st component of displacement:

$$\begin{aligned} u_1(\bar{x}) &= \epsilon_{11}dx_1 + \epsilon_{12}dx_2 + \omega_{11}dx_1 + \omega_{12}dx_2 && (16)\\ &= \frac{\partial u_1}{\partial x_1}dx_1 + \frac{1}{2}(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1})dx_2 + 0dx_1 + \frac{1}{2}(\frac{\partial u_1}{\partial x_2} - \frac{\partial u_2}{\partial x_1})dx_2. && (17) \end{aligned}$$

Likewise the 2nd component is written out as:

$$\begin{aligned} u_2(\bar{x}) &= \epsilon_{21}dx_1 + \epsilon_{22}dx_2 + \omega_{21}dx_1 + \omega_{22}dx_2 && (18)\\ &= \frac{1}{2}(\frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_2})dx_1 + \frac{\partial u_2}{\partial x_2}dx_2 + \frac{1}{2}(\frac{\partial u_2}{\partial x_1} - \frac{\partial u_1}{\partial x_2})dx_1 + 0dx_2. && (19) \end{aligned}$$

This matches our intuition, since we can write out the entire expression as

$$\bar{u}(\bar{x}) = \bar{u}(\bar{x_0}) + \nabla\bar{u}|_{\bar{x_0}} \cdot d\bar{x} \tag{20}$$

from the Taylor expansion.

Problem 2

Consider three material lines $d\bar{x}_a = [1,0]$, $d\bar{x}_b = [1,1]$, and $d\bar{x}_c = [0,1]$, connecting the origin $\bar{x}_0 = [0,0]$ to three other points $\bar{x}_a$, $\bar{x}_b$, and $\bar{x}_c$ that trace the outline of a square with sides of unit length as shown in Figure 1

a. Consider a displacement gradient field such that $\frac{\partial u_1}{\partial x_1} = c$ where c represents some constant value, and all other displacement gradients are zero. Using your results from part (c) above, calculate the two components of displacement occurring at each of the three points $\bar{x}_a$, $\bar{x}_b$, and $\bar{x}_c$ and draw these displacements on the figure. Describe how the shape of a square changes under this particular type of strain

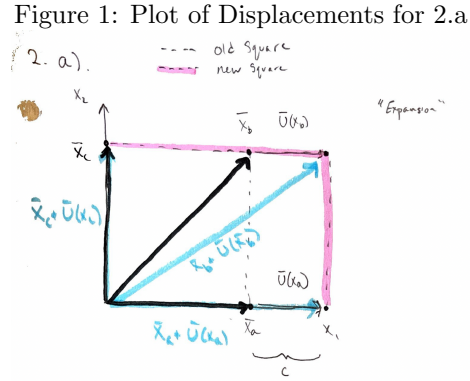
Using our result from 1.C, we can rewrite the displacement in matrix form and solve for both components at once for each material line like:

$$\bar{u}(\bar{x}_a) = \bar{u}(\bar{x}_0) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{21}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} c & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{22}$$

$$= \begin{bmatrix} 0 + c + 0 \\ 0 + 0 + 0 \end{bmatrix} = \begin{bmatrix} c \\ 0 \end{bmatrix}, \tag{23}$$

Figure 1: Plot of Displacements for 2.a



$$\bar{u}(\bar{x_b}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{24}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} c & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{25}$$

$$= \begin{bmatrix} 0 + c + 0 \\ 0 + 0 + 0 \end{bmatrix} = \begin{bmatrix} c \\ 0 \end{bmatrix}, \tag{26}$$

$$\bar{u}(\bar{x_c}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{27}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} c & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{28}$$

$$= \begin{bmatrix} 0 + 0 + 0 \\ 0 + 0 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{29}$$

This results in stretching or compressing the square in the x_1 depending on the sign of c. My sketch of the result is in Figure 1.

b. Repeat with $\frac{\partial u_2}{\partial x_1}$ = c where c represents some constant value, and all other displacement gradients are zero.

Following the procedure in part a, we get:

$$\bar{u}(\bar{x_a}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{30}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{2}c \\ \frac{1}{2}c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{2}c \\ \frac{1}{2}c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{31}$$

$$= \begin{bmatrix} 0 + 00 \\ 0 + \frac{1}{2}c + \frac{1}{2}c \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix}, \tag{32}$$

$$\bar{u}(\bar{x_b}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{33}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{2}c \\ \frac{1}{2}c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{2}c \\ \frac{1}{2}c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{34}$$

$$= \begin{bmatrix} 0 + \frac{1}{2}c - \frac{1}{2}c \\ 0 + \frac{1}{2}c + \frac{1}{2}c \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix}, \tag{35}$$

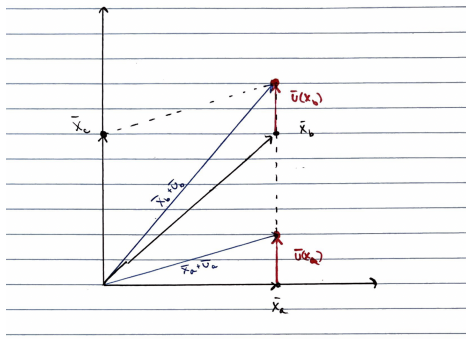
$$\bar{u}(\bar{x_c}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{36}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{2}c \\ \frac{1}{2}c & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{2}c \\ \frac{1}{2}c & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{37}$$

$$= \begin{bmatrix} 0 + \frac{1}{2}c - \frac{1}{2}c \\ 0 + 0 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{38}$$

This results in pulling the right side of the square c units in the x2 direction. This turns the square into a parallelogram. My sketch of this scenario is in Figure 2.

Figure 2: Plot of Displacements for 2.b



c. Repeat with $\frac{\partial u_2}{\partial x_1} = \frac{\partial u_1}{\partial x_2} = c$ where c represents some constant value, and all other displacement gradients are zero.

Following the procedure in part a, we get:

$$\bar{u}(\bar{x_a}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{39}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & c \\ c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{40}$$

$$= \begin{bmatrix} 0+0+0 \\ 0+c+0 \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix}, \tag{41}$$

$$\bar{u}(\bar{x_b}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{42}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & c \\ c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{43}$$

$$= \begin{bmatrix} 0+c+0 \\ 0+c-0 \end{bmatrix} = \begin{bmatrix} c \\ c \end{bmatrix}, \tag{44}$$

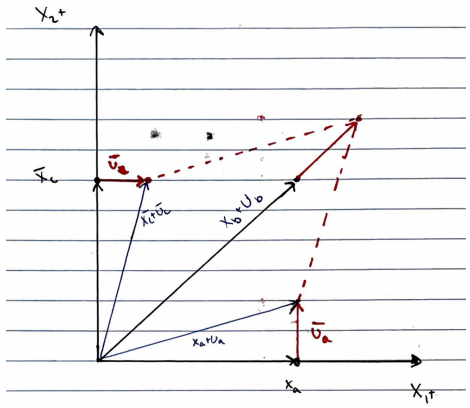
$$\bar{u}(\bar{x_c}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{45}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & c \\ c & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{46}$$

$$= \begin{bmatrix} 0+c+0 \\ 0+0+0 \end{bmatrix} = \begin{bmatrix} c \\ 0 \end{bmatrix}. \tag{47}$$

With these displacement terms, we see a shearing stretch in the x1 and x2 directions. In figure 3, we see that square is being pulled in the diagonal direction, with both sides moving to form a diamond shape.

Figure 3: Plot of Displacements for 2.ac



d. Repeat with $\frac{\partial u_2}{\partial x_1} = c = -\frac{\partial u_1}{\partial x_2} =$ c where c represents some constant value, and all other displacement gradients are zero.

Following the procedure in part a, we get:

$$\bar{u}(\bar{x_a}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{48}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{49}$$

$$= \begin{bmatrix} 0+0+0 \\ 0+0+c \end{bmatrix} = \begin{bmatrix} 0 \\ c \end{bmatrix}, \tag{50}$$

$$\bar{u}(\bar{x_b}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{51}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{52}$$

$$= \begin{bmatrix} 0-c+0 \\ 0+c-0 \end{bmatrix} = \begin{bmatrix} -c \\ c \end{bmatrix}, \tag{53}$$

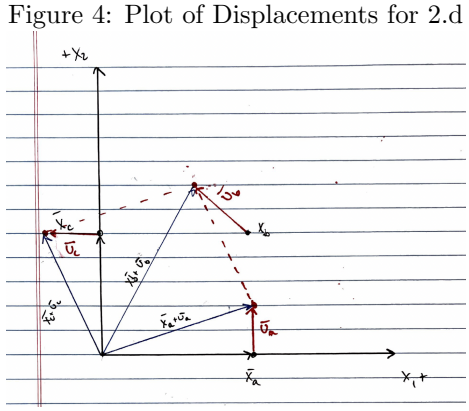
$$\bar{u}(\bar{x_c}) = \bar{u}(\bar{x_0}) + \bar{\bar{\epsilon}} \cdot d\bar{x} + \bar{\bar{\omega}} \cdot d\bar{x} \tag{54}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{55}$$

$$= \begin{bmatrix} 0+-c+0 \\ 0+0+0 \end{bmatrix} = \begin{bmatrix} -c \\ 0 \end{bmatrix}. \tag{56}$$

In this case, we see a pure rotational term, with no strain parts. The square maintains its shape (despite the issues in my sketch) but is rotated about the origin counter clockwise. My sketch is in figure 4.

Figure 4: Plot of Displacements for 2.d



Problem 3

In this exercise we consider the small strains and rotations near a model fault using a solution from elasticity theory. In later weeks we will describe the elastic properties of rock and derive the governing equations for linear elasticity, but for now we focus only on the kinematics (i.e. the displacements) implied by the solution.

Displacement at $|x| \leq a, y = 0^+$, for $\gamma = \Delta\sigma \frac{1-\nu}{G}, \kappa = \Delta\sigma \frac{1-2\nu}{2G}$:

$$u_x = \gamma(a^2 + x^2)^{1/2}, \tag{57}$$
$$u_y = \kappa x. \tag{58}$$

Displacement at $x = 0, y \geq 0$, for $\phi = \Delta\sigma \frac{1}{2G}$:

$$u_x = \gamma[(a^2 + y^2)^{1/2} - y] + \phi[y^2(a^2 + y^2)^{-1/2} - y], \tag{59}$$
$$u_y = 0. \tag{60}$$

a. Evaluate the three small strain components (ϵ_{ij}) at the middle of the model fault where $x = 0, y = 0^+$. Describe each component with reference to material lines at that point and oriented in the two coordinate directions. Explain the magnitude and sign of each component relating these to slip on the model fault and the elastic stiffness of the surrounding rock.

To evaluate ϵ_{ij}, we need to take the partials of each component of u as follows:

$$\frac{\partial u_x}{\partial x}|_{x,y=0} = \frac{\partial}{\partial x}(\gamma(a^2 + x^2)^{1/2})|_{x,y=0} \tag{61}$$
$$= \frac{1}{2}2\gamma x(a^2 + x^2)^{-1/2}|_{x,y=0} \tag{62}$$
$$= \gamma(0)(a^{-1}) \tag{63}$$
$$= 0. \tag{64}$$

$$\frac{\partial u_x}{\partial y}|_{x,y=0} = \frac{\partial}{\partial y}(\gamma[(a^2+y^2)^{1/2} - y] + \phi[y^2(a^2+y^2)^{-1/2} - y])|_{x,y=0} \tag{65}$$

$$= (\gamma[\frac{2y}{2}(a^2+y^2)^{-1/2} - 1] + \phi[2y(a^2+y^2)^{-1/2} + \frac{-(2y)y^2}{2}(a^2+y^2)^{-3/2} - 1])|_{x,y=0} \tag{66}$$

$$= (\gamma[\frac{2(0)}{2}(a^2+(0)^2)^{-1/2} - 1] + \phi[2(0)(a^2+(0)^2)^{-1/2} + \frac{-(2(0))(0)^2}{2}(a^2+(0)^2)^{-3/2} - 1]) \tag{67}$$

$$= -\Delta\sigma\frac{1-\nu}{G} - \Delta\sigma\frac{1}{2G} \tag{68}$$

$$= -\Delta\sigma\frac{1}{2G}(2-2\nu+1) = -\Delta\sigma\frac{1}{2G}(3-\nu). \tag{69}$$

$$\frac{\partial u_y}{\partial x}|_{x,y=0} = \frac{\partial}{\partial x}(\kappa x)|_{x,y=0} \tag{70}$$

$$= \kappa \tag{71}$$

$$= \Delta\sigma\frac{1-2\nu}{2G}. \tag{72}$$

$$\frac{\partial u_y}{\partial y}|_{x,y=0} = \frac{\partial}{\partial y}(0)|_{x,y=0} \tag{73}$$

$$= 0. \tag{74}$$

We can now put all of this to solve for ϵ_{ij} as

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x} = 0, \tag{75}$$

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y} = 0, \tag{76}$$

$$\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2}(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}) \tag{77}$$

$$= \frac{1}{2}(-\Delta\sigma\frac{1}{2G}(3-\nu) + \Delta\sigma\frac{1-2\nu}{2G}) \tag{78}$$

$$= -\frac{\Delta\sigma(2+\nu)}{4G} \tag{79}$$

We can see that there are no normal components of the strain tensor so this case falls into something similar to parts c and d in problem 2. The sign of the off diagonal components is negative so we will see some sort of compression of the material and squeezing into the origin (the opposite as in part c.) Relating these to the model parameters, we can see that the strain will be directly proportional to the magnitude of the slip, and indirectly proportional to the stiffness. So, if the slip increases, so will the shear deformation. Conversely, if the stiffness increases, the strain will decrease in magnitude.

b. Evaluate the small rotation at the middle of the model fault. Describe the rotation with reference to material lines at that point and oriented in the two coordinate directions. Explain the magnitude and sign of the rotation (about the z-axis) relating this to slip on the model fault and the elastic stiffness of the surrounding rock.

Luckily, most of the work was done in part a. We know that the diagonal components of ω_{ij} are 0 since they are symmetric. We can get the other 2 components as:

$$\omega_{xy} = -\omega_{yx} = \frac{1}{2}(\frac{\partial u_x}{\partial y} - \frac{\partial u_y}{\partial x}) \tag{80}$$

$$= \frac{1}{2}(-\Delta\sigma\frac{1}{2G}(3-\nu) - \Delta\sigma\frac{1-2\nu}{2G}) \tag{81}$$

$$= -\frac{\Delta\sigma(4-3\nu)}{4G} \tag{82}$$

Like in part 3.a, the rotation will be directly proportional to slip and indirectly proportional to stiffness. This form looks like our result in part d of problem 2, meaning that the rotation will move in a clockwise fashion about the z axis. This rotation would move the y axis and x axis an equal degrees θ in the counter-clockwise direction about the z axis. Combining this with the result of 3.a, we see a compression and a counter clockwise rotation at x=0, y=0.

c. Use your result from a) and b) to write down the elements of the displacement gradient tensor for the partial derivatives of displacement in terms of a sum of a strain matrix and a rotation matrix. For simplicity let Poisson's ratio be zero.

We can combine our results to get the displacement gradient as:

$$\begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} \end{bmatrix} = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} \\ \epsilon_{yx} & \epsilon_{yy} \end{bmatrix} + \begin{bmatrix} \omega_{xx} & \omega_{xy} \\ \omega_{yx} & \omega_{yy} \end{bmatrix} \tag{83}$$

$$= \begin{bmatrix} 0 & \frac{-\Delta\sigma(2+\nu)}{4G} \\ \frac{-\Delta\sigma(2+\nu)}{4G} & 0 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{\Delta\sigma(4-3\nu)}{4G} \\ \frac{\Delta\sigma(4-3\nu)}{4G} & 0 \end{bmatrix} \tag{84}$$

$$= \begin{bmatrix} 0 & \frac{-\Delta\sigma(3-1\nu)}{2G} \\ \frac{\Delta\sigma(1-2\nu)}{2G} & 0 \end{bmatrix} \tag{85}$$

$$= \begin{bmatrix} 0 & \frac{-\Delta\sigma(3)}{2G} \\ \frac{\Delta\sigma(1)}{2G} & 0 \end{bmatrix}. \tag{86}$$

Luckily, this matches our results in part a.